

2.15 OIL LIMITATIONS

2.15.1 Oil Specifications

	Oil Type	OAT Limit
Engines	MIL-PRF-23699	−30 °C
	MIL-PRF-7808	−30 °C
Main Transmission	MIL-PRF-23699	−30 °C
Intermediate Gearbox	MIL-PRF-23699	−45 °C
Tail Rotor Gearbox	MIL-PRF-23699	−45 °C
Main Rotor Hub	MIL-PRF-23699	−45 °C

2.15.2 Oil Quantities

	Liters	Kilograms
Engines (each tank)	4.5	4.5
Main Transmission	10.3	10.3
Intermediate Gearbox	0.6	0.6
Tail Rotor Gearbox	0.4	0.4
Main Rotor Hub	1.8	1.8
Oil mass values are based on an oil density of 1 kg/l.		

2.16 HYDRAULIC SYSTEM LIMITATIONS

2.16.1 Operational Hydraulic System

For takeoff the operational hydraulic system must be system 1. A change-over to system 2, by operating the hydraulic test switch during flight, is prohibited. The hydraulic test switch is for ground and preflight test purposes only.

2.16.2 Hydraulic Fluid

Hydraulic fluid type MIL-H-5606 is authorized for use at all ambient temperatures.

Hydraulic fluid quantity per system 1.2 liters

2.17 OPERATIONAL LIMITATIONS

2.17.1 Rotor Starting and Stopping in Wind

Rotor starting and stopping has been demonstrated with up to 45 KTAS headwinds.

2.17.2 Hover Turn

Maximum rate 90° per sec

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg*

Maximum rate 45° per sec

EFFECTIVITY ALL

2.17.3 Main Rotor Mast Moment

The mast moment indicator shows the bending moment of the main rotor mast. The red radial marking shall not be exceeded during slope landing, flight maneuvers and ground operation.

Maximum red radial

Caution range yellow arc

Normal range green arc

Calibration point white radial

- NOTE** ● A logbook entry and maintenance action are required whenever:
- the **LIMIT** warning light remains on (even after pointer returns to a lower mast moment), and
 - the helicopter is operated with a defective mast moment system.
- Periodical maintenance actions are required according to Maintenance Manual, if the helicopter is operated without the mast moment system.

2.17.4 Slope Operations

Slope operations (takeoff/landing) are limited in the degree of sloping terrain upon which the maneuver may be performed:

- **Mast moment indicator installed and operative**
 - Ground sloping down to the left Max 15°
 - Ground sloping down to the right Max 8°
 - Ground sloping down forward Max 8°
 - Ground sloping down rearward Max 10°
- **Mast moment indicator inoperative or removed**
 - Ground sloping in all directions Max 5°

2.17.5 Prohibited Flight Maneuvers

Aerobatic maneuvers are prohibited.

2.17.6 Mass-altitude-temperature

For mass-altitude-temperature limitations at takeoff, landing, and during ground effect maneuvers refer to Section 5.

2.17.7 Extreme Low Temperature Operation

Operation at extreme low ambient temperatures (below -30°C) is permitted only under the following conditions:

- Extreme low temperature kit must be installed and operational
- Cockpit/cabin temperatures must be above -30°C
- Engine and transmission oil temperatures must be above 0°C prior to takeoff.

Refer to BO105 Maintenance Manual, Chapter 101, "Measures to be taken when operating at extremely low temperatures".

EFFECTIVITY Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg

Operation at ambient temperatures below -30°C is not permitted.

EFFECTIVITY ALL

2.17.8 Engine Inlet Deflector Shield

EFFECTIVITY 0001-0450 Before SB 60-37

The engine inlet deflector shield must be removed when the OAT is above $+30^{\circ}\text{C}$.

EFFECTIVITY 0451-9999, and 0001-0450 After SB 60-37

The engine inlet deflector shield must be installed at all times.

EFFECTIVITY ALL